

MAT 21D Discussion 2 Notes

Chen Liang

January 15, 2026

§I Warm-Up

Discuss with your group: is the following equation true or false? Why or why not?

$$\int_{-1}^1 \int_{-1}^1 (\sin(xy^2) + \sin(yx^2)) dy dx = 0.$$

Solution. The equation is true. If we try to compute the inner integral, we would find

$$\int_{-1}^1 \sin(yx^2) dy = 0$$

because $\sin(yx^2)$ is an odd function in y . However, we would get stuck at computing

$$\int_{-1}^1 \sin(xy^2) dy.$$

We encounter the same problem even if we apply Fubini's theorem to swap the integral at the beginning. The key to this problem is to apply the additivity of double integrals:

$$\iint_R (f + g) dA = \iint_R f dA + \iint_R g dA.$$

If we let R be the square region $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$, then the integral in question can be rewritten as

$$\iint_R (\sin(xy^2) + \sin(yx^2)) dA = \iint_R \sin(xy^2) dA + \iint_R \sin(yx^2) dA.$$

Then, despite the two double integrals having the same R , we need to tackle them differently, using Fubini's theorem with different orders of integration. For the first double integral, we compute

$$\iint_R \sin(xy^2) dA = \int_{-1}^1 \int_{-1}^1 \sin(xy^2) dx dy = \int_{-1}^1 0 dy = 0,$$

and for the second double integral, we compute

$$\iint_R \sin(yx^2) dA = \int_{-1}^1 \int_{-1}^1 \sin(yx^2) dy dx = \int_{-1}^1 0 dx = 0.$$

§II Average value

The average score times the number of students equals the total score. The average height of a building times the area of the base equals the volume of the buildings. In other words,

$$\text{average score} = \frac{1}{\text{number of students}} \sum_s \text{score of student } s,$$

and

$$\text{average height} = \frac{1}{\text{area of the base}} \iint_R h(x, y) dA.$$

Recall that $\iint_R h(x, y) dA$ represents the volume under the surface $z = h(x, y)$ over the region R .

Recall that the average value of an (integrable) function f over a region R is defined as

$$\text{Average value of } f \text{ over } R = \frac{1}{\text{area of } R} \iint_R f dA.$$

The quantify “area of R ” can be calculated geometrically (either it is a shape that we recognize or can be decomposed into shapes we recognize) or by using the double integral

$$\text{area of } R = \iint_R 1 dA.$$

Let R denote the region of points (x, y) in the plane satisfying $|y| + 1 \leq x$ and $x \leq 3$.

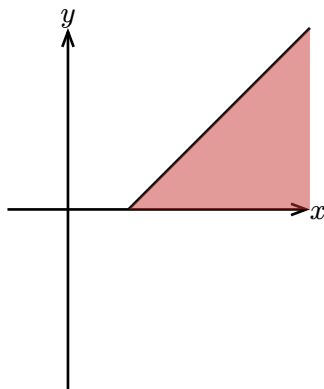
1. Sketch the region R .
2. Calculate the average value of the function $f(x, y) = x$ over the region R . (You can leave your answer as an algebraic expression of fractions.)

Solution.

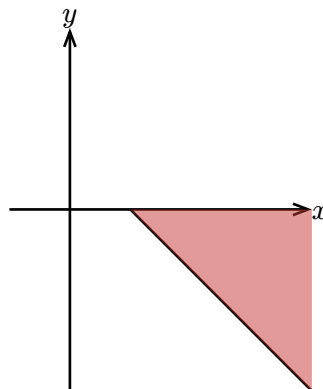
1. To draw an inequality involving absolute values, we need to do caseworks: drawing one region where the value inside the absolute value is positive (or zero) and drawing one region where it is negative. So, for $|y| + 1 \leq x$, we consider two cases:

- Case 1: in the region $y \geq 0$, we need to draw $y + 1 \leq x$.
- Case 2: in the region $y \leq 0$, we need to draw $-y + 1 \leq x$.

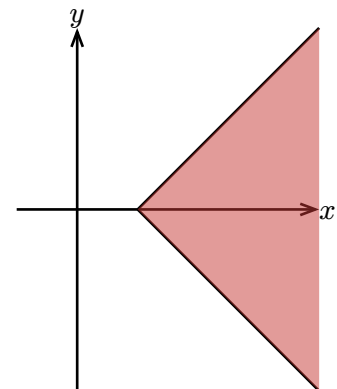
Combining the two cases gives us the $|y| + 1 \leq x$ region.



(a) $y + 1 \leq x$ in the region $y \geq 0$

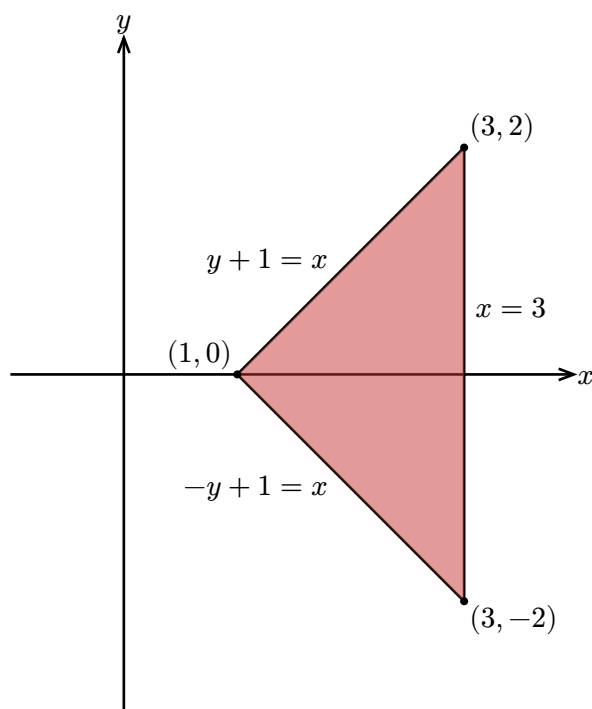


(b) $-y + 1 \leq x$ in the region $y \leq 0$



(c) The region $|y| + 1 \leq x$

Overlapping with $x \leq 3$ gives us the final region R shown below.



2. The region R is a triangle whose area we can calculate geometrically to be 4. Thus, the average value of f over R is

$$\frac{1}{4} \iint_R x \, dA = \frac{1}{4} \int_1^3 \int_{-x+1}^{x-1} x \, dy \, dx.$$

The inner integral is

$$\int_{-x+1}^{x-1} x \, dy = x((x-1) - (-x+1)) = 2x(x-1) = 2x^2 - 2x.$$

Thus,

$$\begin{aligned} \frac{1}{4} \iint_R x \, dA &= \frac{1}{4} \int_1^3 (2x^2 - 2x) \, dx \\ &= \frac{1}{4} \left[\frac{2}{3} x^3 - x^2 \right]_{x=1}^{x=3} \\ &= \frac{1}{4} \left(\frac{2}{3} \cdot 27 - 9 - \left(\frac{2}{3} - 1 \right) \right) \\ &= \frac{7}{3}. \end{aligned}$$

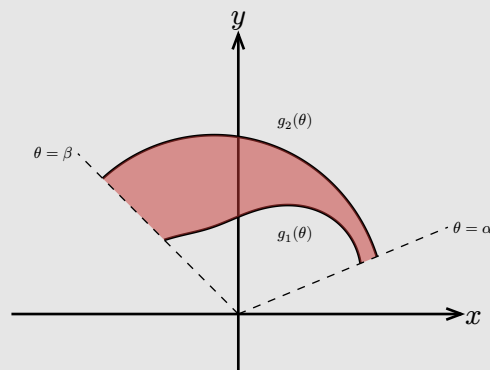
§III Double integrals in polar form

Last week, we learned how to compute double integrals

$$\iint_R f(x, y) \, dA$$

by converting to either $dx \, dy$ or $dy \, dx$ iterated integrals using Fubini's theorem. Today, we will learn another theorem that allows us to compute double integrals.

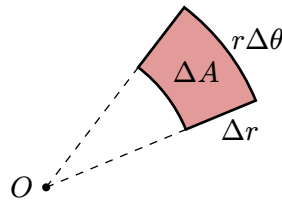
If a region R is given by $\alpha \leq \theta \leq \beta$ and $g_1(\theta) \leq r \leq g_2(\theta)$,



then

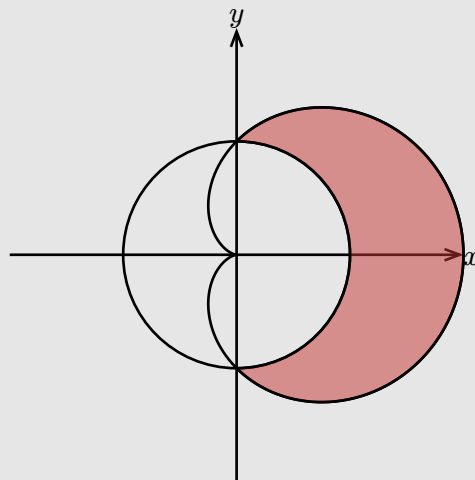
$$\iint_R f(r, \theta) dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} f(r, \theta) r dr d\theta.$$

Pay close attention to the extra factor r . The factor comes from the fact a tiny area element ΔA is approximately a rectangle with side lengths Δr and $r\Delta\theta$.



If you are an scientist or engineer, you may also think in units. The area element dA has unit length squared. The angular element $d\theta$ is unitless, so you need to compensate it by multiplying a length r , so that dr times $r d\theta$ has unit length squared.

The figure below shows two curves: the circle $r = 1$ and the cardioid $r = 1 + \cos(\theta)$. Consider the shaded region R in the figure.



- Express the shaded region R as $\alpha \leq \theta \leq \beta$ and $g_1(\theta) \leq r \leq g_2(\theta)$ for some constants α, β and functions $g_1(\theta), g_2(\theta)$.

2. Express the area of R as an iterated integral in polar coordinates.
3. Calculate the area of R .

Solution.

1. The shaded region R can be expressed as $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ and $1 \leq r \leq 1 + \cos(\theta)$.
2. The area of R can be expressed as

$$\iint_R 1 \, dA = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_1^{1+\cos(\theta)} r \, dr \, d\theta.$$

3. The integral evaluates to

$$\begin{aligned} \iint_R 1 \, dA &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\frac{1}{2} r^2 \right]_{r=1}^{r=1+\cos(\theta)} d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{2} (1 + \cos(\theta))^2 - \frac{1}{2} \right) d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\cos^2(\theta)}{2} + \cos(\theta) \right) d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \cos(2\theta)}{4} + \cos(\theta) \right) d\theta \\ &= \left[\frac{1}{4} \theta + \frac{1}{8} \sin(2\theta) + \sin(\theta) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \left(\frac{1}{4} \left(\frac{\pi}{2} \right) + 0 + 1 \right) - \left(\frac{1}{4} \left(-\frac{\pi}{2} \right) + 0 - 1 \right) \\ &= \frac{\pi}{4} + 2. \end{aligned}$$

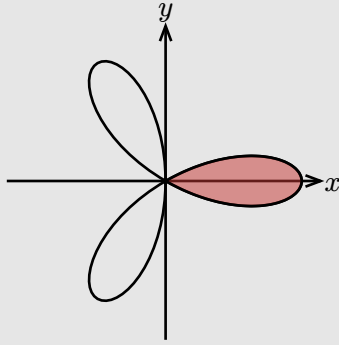
Remark: Note the important identity:

$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}.$$

This is extremely helpful when integrating trigonometric functions. We also have a similar identity for $\sin^2(\theta)$:

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}.$$

The figure below shows the rose $r = 12 \cos(3\theta)$ with the leaf in the right half-plane shaded.



- Express the shaded region R as $\alpha \leq \theta \leq \beta$ and $g_1(\theta) \leq r \leq g_2(\theta)$ for some constants α, β and functions $g_1(\theta), g_2(\theta)$.

Hint: the ray enters and exists the shaded region exactly when it is tangent to the rose.

- Express the area of R as an iterated integral in polar coordinates.
- Find the area of R .

Solution.

- The shaded region R can be expressed as $-\frac{\pi}{6} \leq \theta \leq \frac{\pi}{6}$ and $0 \leq r \leq 12 \cos(3\theta)$. We found $-\frac{\pi}{6}$ and $\frac{\pi}{6}$ because it is exactly at these angles that the ray is tangent to the rose (i.e., $r = 0$).
- The area of R can be expressed as

$$\iint_R 1 \, dA = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \int_0^{12 \cos(3\theta)} r \, dr \, d\theta.$$

- The integral evaluates to

$$\begin{aligned} \iint_R 1 \, dA &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \left[\frac{1}{2} r^2 \right]_{r=0}^{r=12 \cos(3\theta)} d\theta \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 72 \cos^2(3\theta) \, d\theta \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 72 \frac{1 + \cos(6\theta)}{2} \, d\theta \\ &= 36 \left[\theta + \frac{1}{6} \sin(6\theta) \right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \\ &= 36 \left(\left(\frac{\pi}{6} + 0 \right) - \left(-\frac{\pi}{6} + 0 \right) \right) \\ &= 12\pi. \end{aligned}$$